

SERVICENOW PROJECT DOCUMENTATION

IMPORT DATA USING TRANSFORM MAPS (SPREADSHEET)

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Executive Summary

The project demonstrates an end-to-end ServiceNow workflow for importing structured employee data from a spreadsheet. The source workbook is staged through Import Sets, mapped through a Transform Map, transformed into a custom Employee Test table, validated, and then extended with Coalesce, reports, and an analytics dashboard.

Team Details

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1. Ideation Phase

Brainstorming

Project Context

The project focuses on importing structured employee data from an external spreadsheet into ServiceNow. The supplied guide describes a beginner-friendly end-to-end workflow using Import Sets and Transform Maps.

Core Ideas

- Prepare a consistent spreadsheet source.
- Create a dedicated Employee Test target table.
- Use an Import Set as a staging area.
- Map source fields to target fields through a Transform Map.
- Validate transformed records.
- Use Coalesce to handle repeated employee data.
- Create reports and consolidate them in a dashboard.

Idea Relationships

The ideas form a dependency chain. The spreadsheet must exist before loading; the target table must exist before transformation; field mapping must be configured before transformation; validated records are required before meaningful reporting; and the dashboard depends on the reports.

Expected Value

The workflow reduces manual record entry, makes source-to-target mapping explicit, supports repeatable imports, and gives administrators a visual way to review the resulting employee data.

Define Problem Statement

Problem Background

Employee information may arrive in spreadsheet form and may contain multiple rows that need to be inserted or updated in ServiceNow. Manual entry is not a suitable demonstration of bulk migration because it does not show controlled mapping or repeatable transformation.

Problem Statement

The project needs a repeatable mechanism to load spreadsheet-based employee data into ServiceNow, transform it into a structured target table, verify the result, and handle repeated imports without unnecessarily creating duplicate records.

Problem Boundaries

The supplied guide covers spreadsheet preparation, Import Sets, Transform Maps, target-table validation, Coalesce, repeated imports, reports, and dashboards. It does not provide external API integration, custom application code, or quantitative performance measurements.

Success Indicators

Successful transformation, visible records in Employee Test, meaningful Inserted/Updated/Ignored results during repeated imports, successful execution of three reports, and a dashboard containing the reports.

Empathy Map

Primary User

The project is primarily oriented toward a ServiceNow administrator, developer, or learner who needs to migrate employee data and verify the result.

Thinks

The user needs confidence that spreadsheet columns are mapped to the correct ServiceNow fields and that repeated imports will not corrupt or duplicate data.

Does

The user prepares a spreadsheet, creates tables, configures field mappings, runs transformations, checks Transform History, creates reports, and assembles a dashboard.

Feels

A successful workflow should reduce uncertainty by providing visible states such as Completed and Success and by making inserted, updated, and ignored outcomes explicit.

Pains

Manual entry, unclear mappings, duplicate records, and difficulty reviewing large employee datasets can make migration difficult.

Gains

A repeatable import process, controlled field mapping, Coalesce-based matching, and centralized reports provide a clearer workflow and easier verification.

2. Requirement Analysis

Data Flow Diagram and User Stories

End-to-End Data Flow

Spreadsheet → Load Data → Employee Import → Sample Spreadsheet Import → Employee Test → Reports → Employee Analytics Dashboards.

Data Flow Description

The spreadsheet is the source. Load Data places it into the Employee Import staging table. The Transform Map maps the staged source fields to Employee Test. Transformation creates target records, which are then validated and used as the source for reports and dashboard widgets.

User Stories

- As an administrator, I want to upload an Excel spreadsheet so that employee data can be staged in ServiceNow.
- As an administrator, I want to map source fields to target fields so that each value reaches the correct Employee Test column.
- As an administrator, I want to execute a Transform Map so that staged records become target records.
- As an administrator, I want to inspect Transform History so that I can understand inserted, updated, ignored, skipped, and error outcomes.
- As an administrator, I want Coalesce to identify existing records so that repeated imports do not unnecessarily create duplicates.
- As an administrator, I want reports and a dashboard so that employee data can be reviewed visually.

Acceptance Criteria

The source workbook loads successfully, the Transform Map completes successfully, Employee Test contains the expected records, repeated imports demonstrate the documented Coalesce behavior, the three reports run successfully, and the dashboard displays the reports.

Solution Requirements

Functional Requirements

- FR-01: Accept a structured Excel workbook as source data.
- FR-02: Create Employee Test with Employee ID, Employee Name, Email, Department, and Location.
- FR-03: Create Employee Import as the staging Import Set table.
- FR-04: Create Sample Spreadsheet Import with Employee Import as source and Employee Test as target.
- FR-05: Map corresponding source and target fields.
- FR-06: Execute and validate the transformation.
- FR-07: Enable Coalesce on a field map.

- FR-08: Review repeated-import results using Transform History.
- FR-09: Create department, location, and list reports.
- FR-10: Add all three reports to Employee Analytics Dashboards.

Non-Functional Requirements

The guide demonstrates consistency, traceability, data integrity, usability of the import workflow, and clear validation evidence. It does not specify numeric performance targets.

Data Requirements

The source uses Employee ID, Name, Email, Department, and Location. The target uses Employee ID, Employee Name, Email, Department, and Location.

Security / Access Consideration

The guide includes dashboard sharing controls for Users, Groups, and Roles. No additional security architecture is documented.

Technology Stack

Source Technology

Google Spreadsheet is used to prepare the sample data, which is downloaded in Microsoft Excel (.xlsx) format.

ServiceNow Data Layer

The Employee Test table stores transformed employee records; Employee Import provides the Import Set staging table.

Transformation Layer

The Sample Spreadsheet Import Transform Map provides source-to-target mapping and transformation.

Data Integrity Layer

Coalesce and Transform History support repeated-import handling and result inspection.

Presentation Layer

ServiceNow Reports and Employee Analytics Dashboards present the imported data.

Project Boundary

No custom programming language, external database, external API, or custom backend is specified in the uploaded guide.

3. Project Design Phase

Problem – Solution Fit

Problem

Employee information is supplied as spreadsheet data and must be migrated into ServiceNow without manually recreating every record.

Solution

Use Import Sets to stage the workbook, a Transform Map to map source fields to Employee Test, Coalesce to identify existing records during repeated imports, and reports/dashboard views for verification.

Fit Analysis

Bulk data movement

Spreadsheet + Import Set supports loading multiple records.

Correct mapping

Transform Map and Field Maps make source-to-target relationships explicit.

Repeated imports

Coalesce supports matching existing records.

Verification

Transform History and Employee Test provide visible validation.

Visibility

Reports and Dashboard provide summarized views.

Expected Outcome

The design supports a repeatable and observable path from external spreadsheet to ServiceNow records and analytics.

Proposed Solution

Workflow

1) Prepare spreadsheet; 2) create Employee Test; 3) create Employee Import; 4) create Sample Spreadsheet Import; 5) map fields; 6) transform; 7) validate; 8) enable Coalesce; 9) repeat import; 10) create reports; 11) build dashboard.

Target Data Model

Employee Test contains Employee ID, Employee Name, Email, Department, and Location as String fields.

Transformation Logic

The Transform Map connects Employee Import to Employee Test. Auto Map Matching Fields and Mapping Assist are used to establish source and target field relationships.

Repeat Import Logic

Coalesce is enabled on a field map. The guide then demonstrates new IDs being inserted and existing IDs with changed values being updated. Repeating the same sheet results in ignored rows according to the documented result.

Analytics

Three reports are created and added to Employee Analytics Dashboards.

Solution Architecture

Architecture Layers

Source

Google Spreadsheet / Excel

Staging

Employee Import

Transformation

Sample Spreadsheet Import

Target

Employee Test

Integrity

Coalesce / Transform History

Presentation

Reports / Dashboard

Flow

Spreadsheet → Employee Import → Sample Spreadsheet Import → Employee Test → Reports → Employee Analytics Dashboards.

Design Rationale

The architecture separates incoming data from the target table. This makes mapping explicit and allows the project to inspect transformation results before relying on the target records for reporting.

4. Project Planning Phase

Planning Logic

Dependency Logic

Spreadsheet preparation precedes import-table loading. The target table must exist before the Transform Map can point to it. Field mapping precedes transformation. Transformation and validation precede reporting. Reports precede dashboard integration.

Work Packages

- WP1 – Prepare source workbook.
- WP2 – Create Employee Test and fields.
- WP3 – Create Employee Import.
- WP4 – Configure Sample Spreadsheet Import.
- WP5 – Transform and validate.
- WP6 – Configure Coalesce and repeat import.
- WP7 – Create reports.
- WP8 – Create and populate dashboard.

Risk and Mitigation

Incorrect mapping

Review Field Maps before transformation.

Duplicate data

Use Coalesce and inspect Transform History.

Unverified result

Open Employee Test and arrange list columns.

Dashboard incomplete

Add and verify all three reports.

Completion Logic

Each work package is considered complete when the corresponding configuration or result is visible in ServiceNow, as represented by the source screenshots.

Project Planning

Implementation Sequence

The project is planned as a sequential workflow rather than independent activities because each later stage depends on the output of the previous stage.

Milestone Plan

Milestone 1

Spreadsheet and Employee Test target table.

Milestone 2

Employee Import and Transform Map.

Milestone 3

Transformation, validation, Coalesce, repeated import.

Milestone 4

Reports and Employee Analytics Dashboard.

Deliverables

The final implementation produces a populated Employee Test table, a configured Transform Map, repeat-import evidence, three reports, and a dashboard.

Documentation Plan

Screenshots are retained as evidence for table creation, import configuration, mapping, transformation, repeated-import results, reports, and dashboard configuration.

5. Project Development Phase

Performance Testing

Testing Scope

The supplied guide does not contain numerical load or performance benchmarks. The development evidence focuses on operational correctness of the import and transformation workflow.

Functional Performance Indicators

The guide checks Completed and Success states for data loading and transformation, successful record creation, and clear Transform History outcomes.

Repeated Import Observation

The project demonstrates four rows being uploaded, with two new IDs inserted and two existing IDs updated in one scenario. Repeating the same sheet is documented with four ignored rows.

Limitation

No response-time, throughput, concurrent-user, CPU, memory, or scalability values are provided in the source. Such values should not be presented as measured project results.

UAT Report

UAT Objective

Confirm that the documented business workflow can prepare, import, transform, validate, update, and visualize employee records.

Acceptance Scenarios

UAT-01

Source spreadsheet can be prepared and downloaded.

UAT-02

Employee Test fields are available.

UAT-03

Employee Import can stage the workbook.

UAT-04

Sample Spreadsheet Import transforms data successfully.

UAT-05

Employee Test displays imported records.

UAT-06

Coalesce supports repeated-import behavior.

UAT-07

Three reports and the dashboard display the resulting data.

Acceptance Interpretation

The source screenshots provide evidence for the listed scenarios. Formal UAT signatures, external business-owner approvals, and defect statistics are not provided.

UAT Execution Report

Execution Method

Execute each scenario in the same order as the project guide and capture the ServiceNow state or screen that demonstrates the result.

Expected Evidence

Load Data success, Transform success, Employee Test records, Coalesce field-map configuration, Transform History, report output, and dashboard output.

Observed Source Evidence

The guide documents successful import and transformation, visible target records, inserted/updated/ignored repeated-import results, and successful report/dashboard configuration.

Execution Limitation

The uploaded guide does not provide formal tester identity, execution date, defect ID, or sign-off fields. These are therefore left as source limitations.

6. Project Documentation

Functional Specification Document

Purpose

Define the functional behavior of the spreadsheet-to-ServiceNow employee data import solution.

Inputs

Excel workbook containing Employee ID, Name, Email, Department, and Location.

Processing

Load Data stages the workbook in Employee Import; Sample Spreadsheet Import maps source fields to Employee Test; transformation writes target records.

Data Integrity

Coalesce is enabled on a field map to support repeated imports and matching of existing records.

Outputs

Employee Test records, Transform History results, Employees by Department report, Employees by Location report, Employee List Report, and Employee Analytics Dashboards.

Constraints

No custom code, external API, external database, or measured performance target is documented.

Final Report

Executive Summary

This project implements a ServiceNow employee-data import workflow using spreadsheet data, Import Sets, Transform Maps, Coalesce, Reports, and Dashboards.

Implementation Summary

The source workbook is staged in Employee Import and transformed into Employee Test using Sample Spreadsheet Import. The target records are validated and then used for analytics.

Coalesce Enhancement

The guide demonstrates how existing employee IDs can be updated rather than duplicated and how repeating the same sheet can result in ignored rows.

Reporting Enhancement

Employees by Department, Employees by Location, and Employee List Report provide different views of the target data.

Dashboard

Employee Analytics Dashboards consolidates the reports and includes sharing controls.

Conclusion

The source concludes that the solution supports accuracy, efficiency, data integrity, and centralized visibility.

7. Project Demonstration

Project Demonstration

Demo Sequence

Begin with the spreadsheet, then show Employee Test, Employee Import, Sample Spreadsheet Import, field mappings, Transform success, Employee Test validation, Coalesce, repeated import results, reports, and dashboard.

Talking Points

Explain that Import Sets provide staging, Transform Maps provide source-to-target mapping, Coalesce supports matching during repeated imports, and reports/dashboard provide visibility.

Evidence

Use the milestone screenshots in the corresponding folders to support each stage.

Final Demonstration Result

The complete demonstration should show a continuous workflow from external spreadsheet to ServiceNow target records and analytics.

Integrated Project Workflow

Stage	Configuration	Outcome
1	Spreadsheet preparation	Excel source ready.
2	Employee Test / u_employee_test	Target schema created.
3	Employee Import / u_employee_import	Source staged.
4	Sample Spreadsheet Import	Fields mapped and transformation configured.
5	Transform + validation	Records created and verified.
6	Coalesce + repeat import	Existing records updated; repeated identical rows ignored according to guide.
7	Three reports	Department, Location, and Employee List views.
8	Employee Analytics Dashboards	Reports consolidated for centralized visibility.

Conclusion

The project provides a clear demonstration of spreadsheet-driven ServiceNow data migration. Its strongest implementation features are explicit source-to-target mapping, validation through Transform History and Employee Test, repeated-import handling through Coalesce, and presentation through reports and a dashboard.